PRACTICAL - 2

OBJECTIVE:

The objective of this practical is to learn how to install and configure Hadoop in a Single-Node Cluster environment. It also helps in understanding HDFS operations such as storing, managing, retrieving, and monitoring big data efficiently.

AIM: Hadoop Installation, Configuration and HDFS Operations on Windows/Linux

Problem Definition

You have recently joined DataAnalytics Inc. as a Hadoop Administrator. The organization processes large-scale customer, sales, and marketing data using the Hadoop ecosystem. Due to increasing data volumes, the company has decided to migrate its data storage infrastructure from traditional file systems to Hadoop Distributed File System (HDFS). As a Hadoop Administrator, your responsibility is to install and configure Hadoop in a Single-Node Cluster environment on Windows/Linux, verify the Hadoop services, manage HDFS storage, and perform data management operations. Management wants a fully functional Hadoop environment that can store, retrieve, and process big data efficiently. Your task is to install Hadoop, configure the Hadoop ecosystem, start Hadoop services, and perform HDFS operations using Hadoop commands.

Task:

- **Hadoop Installation & Configuration**
- **Access Hadoop Web UI.**

1. **Create directories in HDFS.**
2. **Upload files to HDFS.**
3. **List HDFS files and directories.**
4. **Display file contents.**
5. **Copy files within HDFS.**
6. **Move files within HDFS.**
7. **Rename files.**
8. **Download files from HDFS.**
9. **Delete files and directories.**
10. **Check HDFS report.**
11. **Verify block information.**

12. Check Hadoop cluster health.

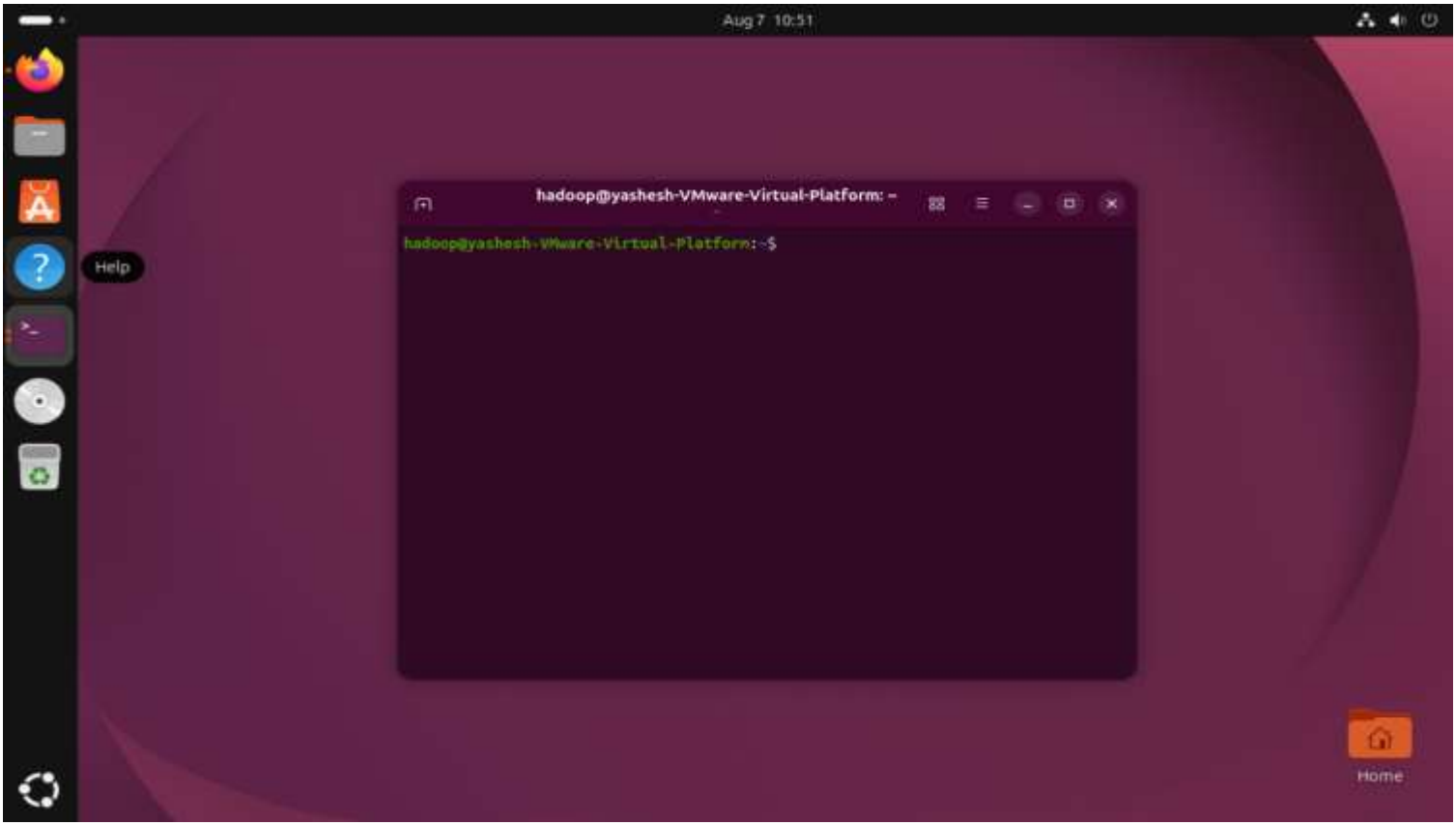
13. Monitor Hadoop processes.

14. Check storage utilization.

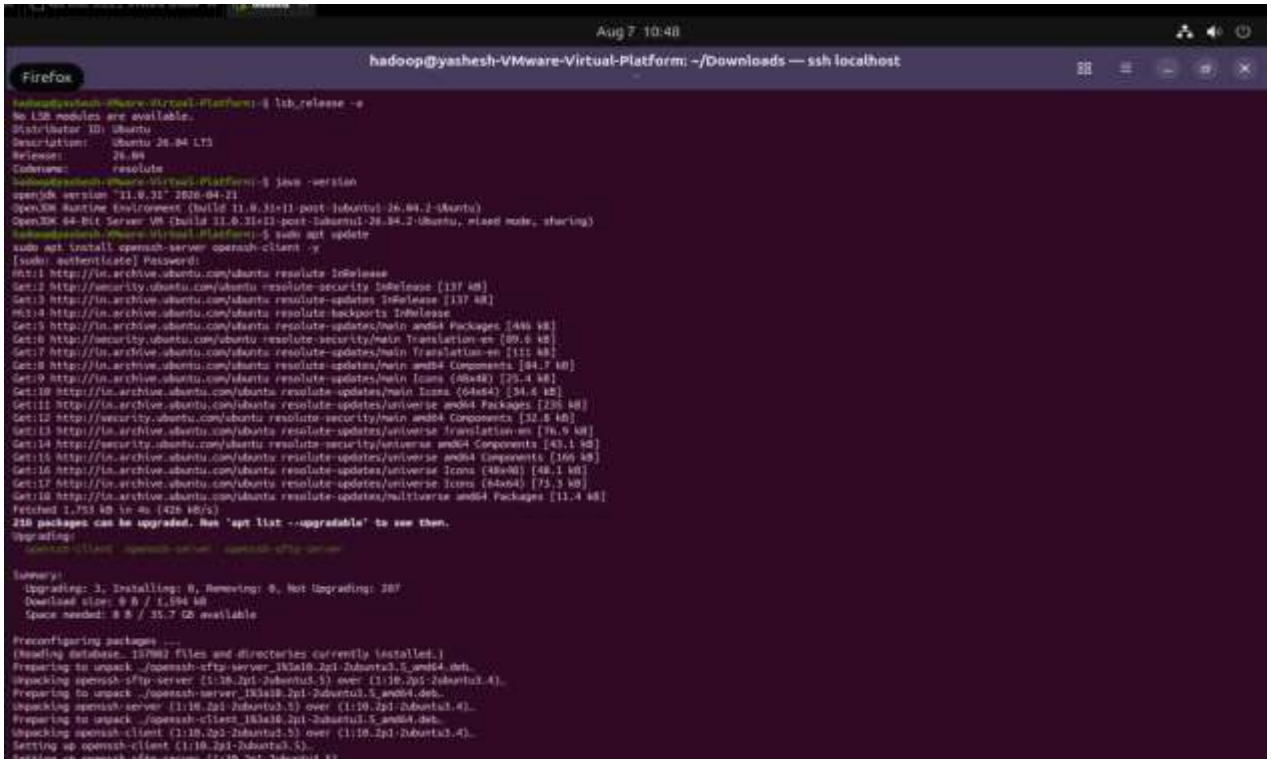
Tools / Technology Used

- Windows 10 / Windows 11
- OpenJDK 11 or above
- Hadoop 3.x
- Winutils.exe
- Command Prompt / PowerShell

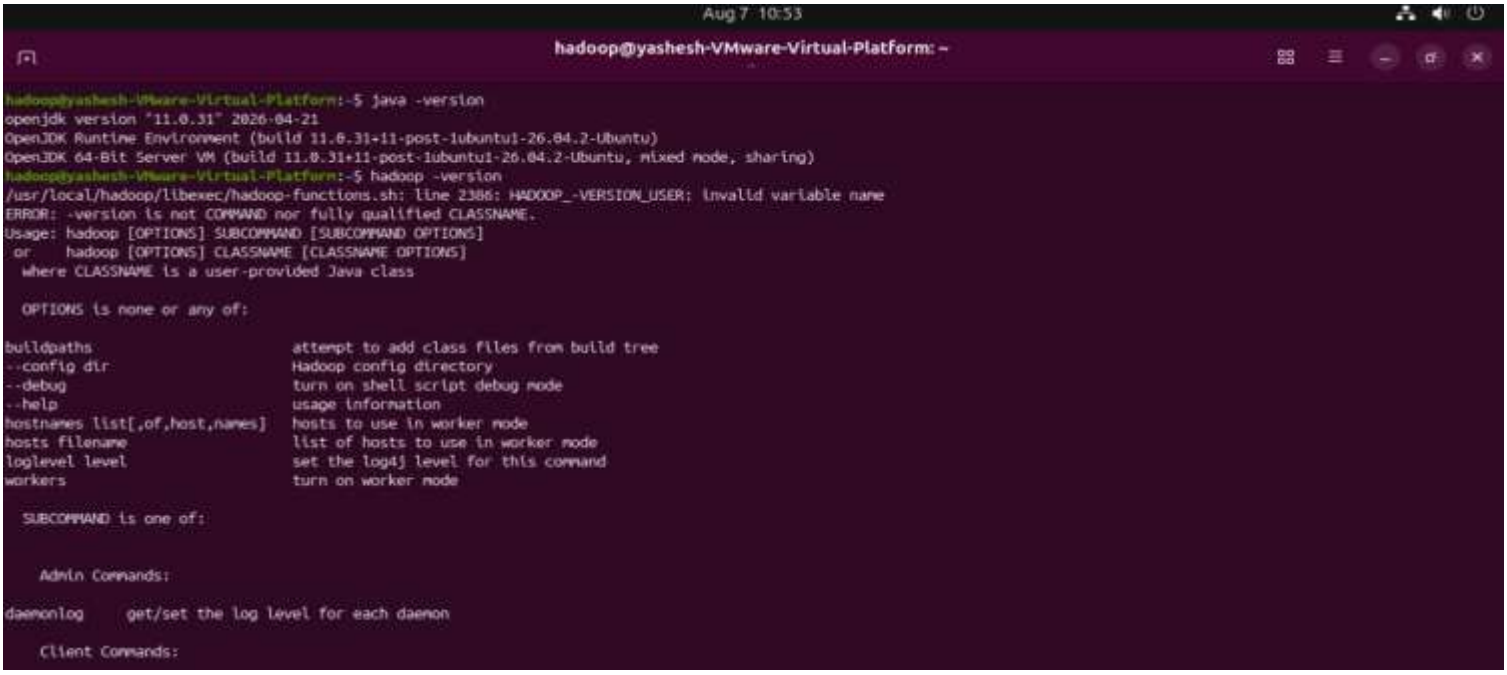
Step 1: Open Terminal



Step 2: Check Java



Step 3: Check Hadoop Version



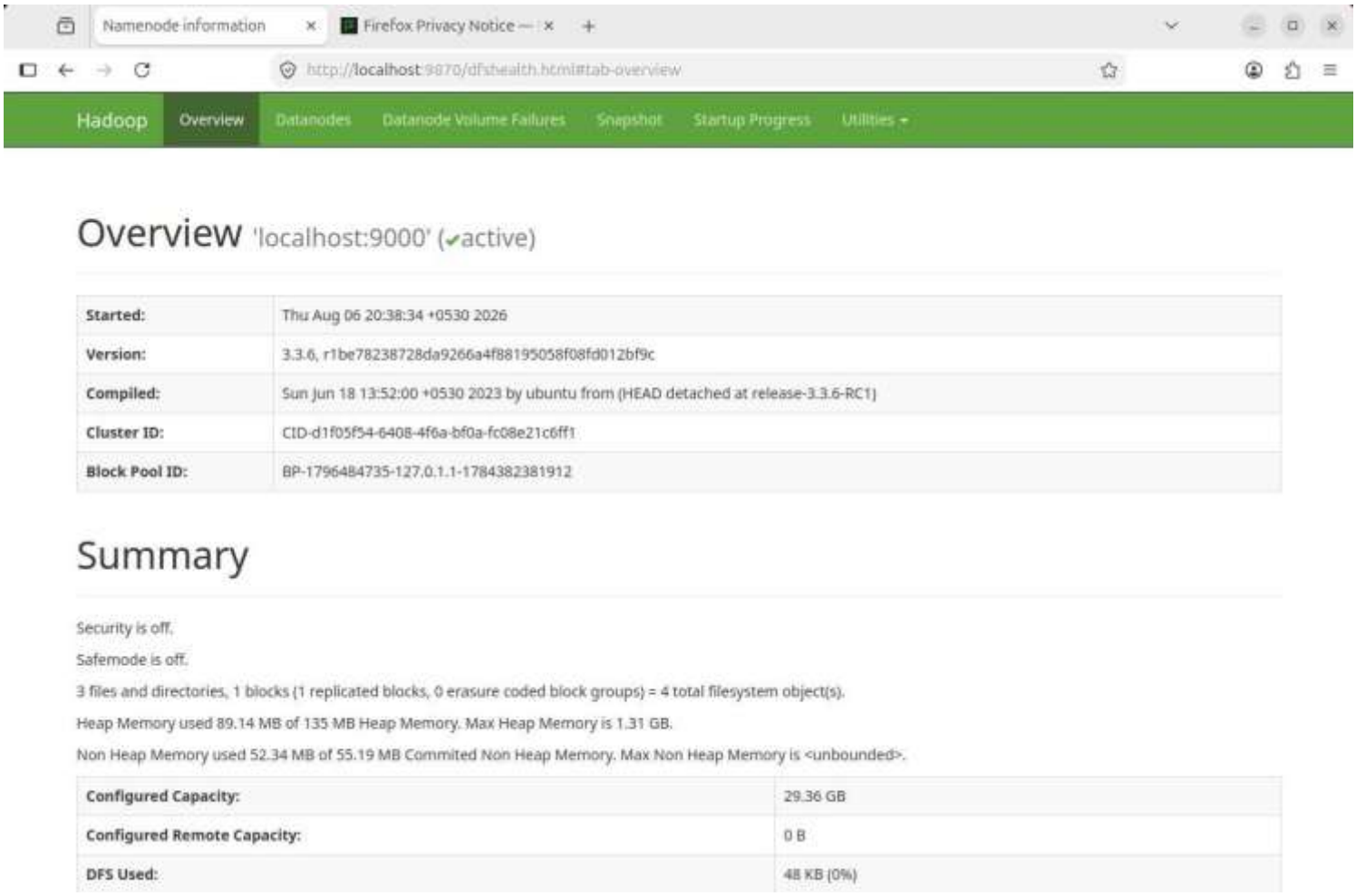
Step 4: Start Hadoop



Step 5: Verify Services



Access Hadoop Web UI



Step 6: Create Dataset

```
hadoop@yashesh-VMware-Virtual-Platform:~$ nano sales_data.txt
hadoop@yashesh-VMware-Virtual-Platform:~$ cat sales_data.txt
ProductA 120
ProductB 80
ProductC 150
ProductA 200
ProductB 90
```

Practical Commands

1 Create Directory

```
hadoop@yashesh-VMware-Virtual-Platform:~/Downloads$ hdfs dfs -mkdir /student
2026-08-07 10:25:55,159 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
hadoop@yashesh-VMware-Virtual-Platform:~/Downloads$ hdfs dfs -mkdir /student/files
2026-08-07 10:26:07,674 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
hadoop@yashesh-VMware-Virtual-Platform:~/Downloads$ hdfs dfs -ls /
2026-08-07 10:26:16,588 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Found 3 items
drwxr-xr-x   - hadoop supergroup          0 2026-08-07 10:20 /mydir
drwxr-xr-x   - hadoop supergroup          0 2026-08-07 10:26 /student
drwxr-xr-x   - hadoop supergroup          0 2026-08-07 10:19 /test2
```

2 Upload File

```
hadoop@yashesh-VMware-Virtual-Platform:~$ echo "Hello Hadoop Practical" > sample.txt
hadoop@yashesh-VMware-Virtual-Platform:~$ hdfs dfs -put sample.txt /student
2026-08-07 14:52:41,670 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applica
ble
hadoop@yashesh-VMware-Virtual-Platform:~$ hdfs dfs -ls /student
2026-08-07 14:52:50,128 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applica
ble
Found 2 items
drwxr-xr-x   - hadoop supergroup          0 2026-08-07 14:51 /student/files
-rw-r--r--   1 hadoop supergroup         23 2026-08-07 14:52 /student/sample.txt
```

3 List Files

```
hadoop@yashesh-VMware-Virtual-Platform:~$ hdfs dfs -ls /student
2026-08-07 14:53:47,515 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applica
ble
Found 2 items
drwxr-xr-x   - hadoop supergroup          0 2026-08-07 14:51 /student/files
-rw-r--r--   1 hadoop supergroup         23 2026-08-07 14:52 /student/sample.txt
```

4 Display File

```
hadoop@yashesh-VMware-Virtual-Platform:~$ hdfs dfs -cat /student/sample.txt
2026-08-07 14:54:55,872 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applica
ble
Hello Hadoop Practical
```

5 Copy File

```
hadoop@yashesh-VMware-Virtual-Platform:~$ hdfs dfs -cp /student/sample.txt /student/sample_copy.txt
2026-08-07 14:55:44,385 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applica
ble
hadoop@yashesh-VMware-Virtual-Platform:~$ hdfs dfs -ls /student
2026-08-07 14:56:02,824 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applica
ble
Found 3 items
drwxr-xr-x   - hadoop supergroup          0 2026-08-07 14:51 /student/files
-rw-r--r--   1 hadoop supergroup         23 2026-08-07 14:52 /student/sample.txt
-rw-r--r--   1 hadoop supergroup         23 2026-08-07 14:55 /student/sample_copy.txt
```

6 Move File

```
hadoop@yashesh-VMware-Virtual-Platform:~$ hdfs dfs -mv /student/sample_copy.txt /student/files/
2026-08-07 14:56:47,442 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applica
ble
hadoop@yashesh-VMware-Virtual-Platform:~$ hdfs dfs -ls /student/files
2026-08-07 14:56:56,876 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applica
ble
Found 1 items
-rw-r--r--   1 hadoop supergroup         23 2026-08-07 14:55 /student/files/sample_copy.txt
```

7 Rename File

```
hadoop@yashesh-VMware-Virtual-Platform:~$ hdfs dfs -mv /student/sample.txt /student/renamed.txt
2026-08-07 14:57:55,357 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applica
ble
hadoop@yashesh-VMware-Virtual-Platform:~$ hdfs dfs -ls /student
2026-08-07 14:58:08,480 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applica
ble
Found 2 items
drwxr-xr-x   - hadoop supergroup          0 2026-08-07 14:56 /student/files
-rw-r--r--   1 hadoop supergroup         23 2026-08-07 14:52 /student/renamed.txt
```

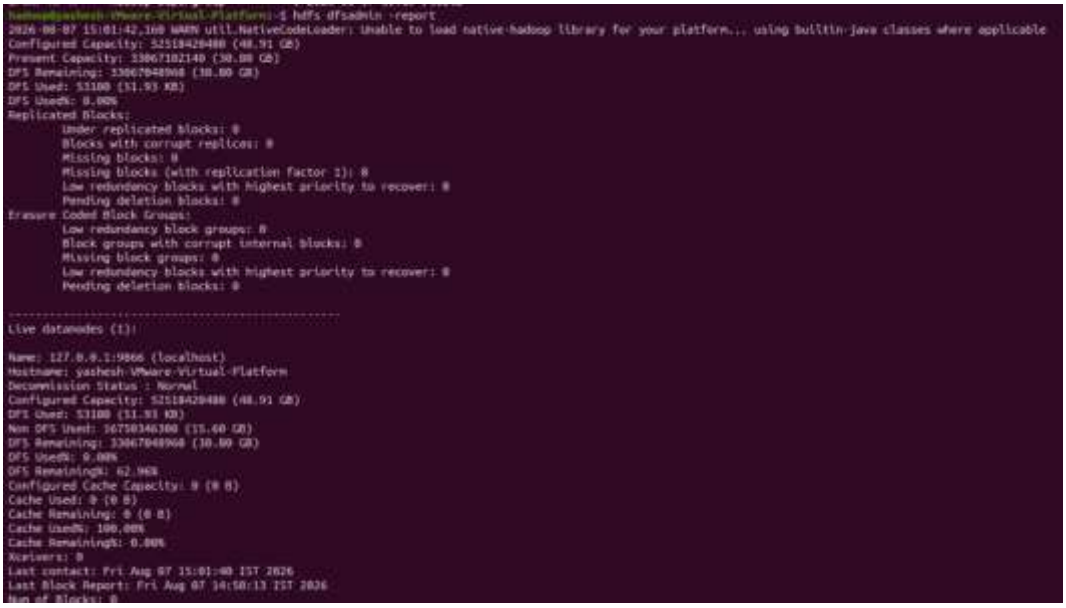
8 Download File

```
hadoop@yashesh-VMware-Virtual-Platform:~$ mkdir downloaded
hadoop@yashesh-VMware-Virtual-Platform:~$ hdfs dfs -get /student/renamed.txt downloaded/
2026-08-07 14:58:54,486 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applica
ble
hadoop@yashesh-VMware-Virtual-Platform:~$ ls downloaded
renamed.txt
hadoop@yashesh-VMware-Virtual-Platform:~$ cat downloaded/renamed.txt
Hello Hadoop Practical
```

9 Delete File

```
hadoop@yashesh-VMware-Virtual-Platform:~$ hdfs dfs -rm /student/renamed.txt
2026-08-07 15:00:04,709 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applica
ble
Deleted /student/renamed.txt
hadoop@yashesh-VMware-Virtual-Platform:~$ hdfs dfs -rm -r /student/files
2026-08-07 15:00:22,478 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applica
ble
Deleted /student/files
hadoop@yashesh-VMware-Virtual-Platform:~$ hdfs dfs -rm -r /student
2026-08-07 15:00:38,108 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applica
ble
Deleted /student
hadoop@yashesh-VMware-Virtual-Platform:~$ hdfs dfs -ls /
2026-08-07 15:00:48,044 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applica
ble
Found 2 items
drwxr-xr-x   - hadoop supergroup          0 2026-08-07 10:20 /mydir
drwxr-xr-x   - hadoop supergroup          0 2026-08-07 10:19 /test2
```

10 HDFS Report



11 Verify Block Information



12 Check Cluster Health



13 Monitor Hadoop Processes

```
hadoop@yashesh-VMware-Virtual-Platform:~$ jps
4736 NameNode
6936 Jps
4906 DataNode
5102 SecondaryNameNode
```

14 Storage Utilization

```
hadoop@yashesh-VMware-Virtual-Platform:~$ hdfs dfs -df -h
2026-08-07 15:07:21,376 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java clas
ses where applicable
Filesystem              Size    Used  Available  Use%
hdfs://localhost:9000  48.9 G  51.9 K     30.8 G    0%
```

15 Stop Hadoop

```
hadoop@yashesh-VMware-Virtual-Platform:~$ stop-yarn.sh
Stopping nodemanagers
Stopping resourcemanager
hadoop@yashesh-VMware-Virtual-Platform:~$ stop-dfs.sh
Stopping namenodes on [localhost]
Stopping datanodes
Stopping secondary namenodes [yashesh-VMware-Virtual-Platform]
2026-08-07 15:08:39,230 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java clas
ses where applicable
```

Key Questions / Analysis

Q1. Why is Hadoop preferred for Big Data storage?

Hadoop can store and process very large amounts of data across multiple computers. It is fast, reliable, and cost-effective.

Q2. What is the role of NameNode and DataNode in HDFS?

The NameNode manages file information and keeps track of where data is stored. The DataNode stores the actual data blocks and sends them when requested.

Q3. What is the difference between HDFS and traditional file systems?

HDFS stores data across multiple machines, while a traditional file system stores data on a single computer. HDFS is designed for handle.

Q4. What are the advantages of distributed storage?

Distributed storage improves speed, reliability, scalability, and allows data to remain available even if one machine fails.

Supplementary Questions

1. Configure Hadoop pseudo-distributed mode.

Pseudo-distributed mode allows Hadoop services to run on a single computer for learning, testing, and development.

---

2. Change HDFS replication factor and observe results.

Changing the replication factor changes the number of copies of each data block stored in HDFS, improving data availability and fault tolerance.

---

3. Upload a large dataset into HDFS.

A large dataset is uploaded into HDFS using Hadoop commands so it can be stored and processed efficiently.

---

4. Analyze HDFS storage utilization.

Storage utilization is checked to understand how much HDFS space is used and how much is available.

Conclusion

In this practical, Hadoop was successfully installed and configured in a Single-Node Cluster environment. HDFS operations such as creating directories, uploading, downloading, moving, renaming, and deleting files were performed successfully. Hadoop services and storage utilization were also monitored. This practical helped in understanding how Hadoop manages and stores large amounts of data efficiently using HDFS.

Key Skills to be Addressed

- Hadoop Installation and Configuration
- Environment Variable Configuration
- HDFS Administration
- Distributed Storage Management
- Hadoop Cluster Monitoring

Dataset / TestData

sales_data.txt
ProductA 120
ProductB 80
ProductC 150
ProductA 200
ProductB 90
Format
TXT/CSV Files (mb/gb)